EE 451: Communications Systems

Detailed Lesson-by-Lesson Schedule - Spring 2026

Class Meeting Times: Tuesday & Thursday, 2:30 PM - 3:45 PM, Loyola Science Center Room 142 **Textbook:** An Introduction to Analog and Digital Communications (2nd Ed.), Haykin & Moher

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Reading Quiz Schedule

Reading quizzes administered through Brightspace at the beginning of selected classes:

Lesson	Date	Topic	Reading
2	Tue, Feb 3	Fourier Transform, FT Properties, Dirac Delta, Periodic Signals	Chapter 2.1-2.5
4	Tue, Feb 10	Amplitude Modulation Techniques	Chapter 3.1-3.3
7	Thu, Feb 19	SSB, VSB, Receiver Architectures, ASK/OOK	Chapter 3.6-3.8, 7.1-7.2
11	Thu, Mar 5	FM/PM Theory and Modulation	Chapter 4.1-4.5
13	Thu, Mar 12	FM Generation/Demodulation, FSK/BPSK	Chapter 4.6-4.8, 7.3-7.4
18	Thu, Apr 9	Noise in Communications Systems	Chapter 9
22	Thu, Apr 23	Probability and Random Variables	Chapter 8.1-8.2
25	Tue, May 5	Random Variables, Noise in Analog Communications	Chapter 8.3-8.4, 9
27	Tue, May 12	Digital Performance, BER, System Noise Calculations	Chapter 10, 11.1-11.3

Homework Schedule

Assignment	Assigned	Due	Topics
Homework 1	Thu, Jan 29	Thu, Feb 5	Complex numbers, phasors, signals
Homework 2	Thu, Feb 12	Thu, Feb 19	AM and ASK
Homework 3	Thu, Mar 5	Tue, Mar 24	FM, PM, FSK, BPSK
Homework 4	Thu, Apr 16	Thu, Apr 23	M-ary modulation, QAM, EVM
Homework 5	Thu, Apr 23	Thu, May 7	Probability, noise, SNR

Exam Schedule

Exam	Date	Coverage
Exam 1	Thu, Feb 26	Chapters 2-3, 7.1-7.2 (Fourier Analysis & AM)
Exam 2	Tue, Apr 21	Chapters 4-7, 9, 11 (Modulation, Noise & Link Budgets)
Final Exam	Thu, May 21	Comprehensive (emphasis on Chapters 8-11)

Lab Schedule

Python/Jupyter Labs

Lab	Week	Date	Topic
Python Lab 1	2	Thu, Feb 5	Fourier Analysis and Spectral Visualization
Python Lab 2	4	Thu, Feb 19	AM/ASK Modulation and Envelope Detection
Python Lab 3	11	Tue, Apr 14	CDMA & Spread Spectrum Simulation
Python Lab 4	11	Thu, Apr 16	QPSK/QAM Simulation with EVM Analysis
Python Lab 5	15	Tue, May 12	BER Performance Simulation

GNU Radio Labs

Lab	Week	Date	Topic
GNU Radio Lab 1	10	Thu, Apr 9	FM Broadcast Reception & WiFi Spectrum Analysis
GNU Radio Lab 2	14	Tue, May 5	Noise Analysis and SNR Measurement

Baba Yaga's Hut Phasor Labs

Session	Week	Date	Topic
Part 1	8	Thu, Mar 26	AM Phasor Analysis with I/Q Demodulation
Part 2	9	Tue, Mar 31	DSB-SC, FM, and PM Phasor Analysis

W3USR Amateur Radio Station Activities

Activity	Week	Date	Topic
W3USR Activity 1	4	Thu, Feb 19	HF Station Tour and SSB/AM Reception
W3USR Activity 2	11	Thu, Apr 16	Digital Modes (FT8, APRS, PSK31)

Optional Extra Credit: W3USR Satellite Communications activity available by arrangement (contact instructor)

Phase 1: Foundation (Weeks 1-2)

Week 1

Lesson 1 - Thursday, January 29 Course Introduction & Complex Signal Review - Course overview, syllabus, grading policy, amateur radio extra credit opportunities - Introduction to W3USR amateur radio station capabilities - Cross-disciplinary applications: radar, sonar, audio processing, biomedical - Complex numbers and Euler's formula review - Sinusoids: amplitude, frequency, phase - Reading: Chapter 1, Chapter 2.1-2.5 (in preparation for Reading Quiz 1 in Lesson 2)

Assignment: Install Python/Jupyter environment before next class (see environment.yml)

Homework 1 Assigned: Complex numbers, phasors, basic signal operations

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Week 2

Lesson 2 - Tuesday, February 3 Signal Analysis Fundamentals & I/Q Representation - I/Q (In-phase/Quadrature) representation - industry-critical terminology - Convolution, unit step and impulse functions - Energy and power signals - Phasor representation of sinusoids - Introduction to analytic signals

READING QUIZ 1 - Topic: Fourier Transform, FT Properties, Dirac Delta, Periodic Signals - Reading: Chapter 2.1-2.5 - Administered via Brightspace at beginning of class

Note: Correlation, a related operation to convolution, will be introduced in Lesson 4 after we study Fourier transforms. Correlation is essential for matched filtering and signal detection in communication systems.

Lesson 3 - Thursday, February 5 Fourier Analysis Essentials - Fourier series and Fourier transforms - Properties of Fourier transforms - Time-frequency duality - Bandwidth concepts - Reading: Chapter 3.1-3.3 (in preparation for Reading Quiz 2 in Lesson 4)

Lab: Python Lab 1 - Fourier Analysis - Generate and plot sinusoids - Compute and visualize Fourier transforms - Explore time-frequency duality - Bandwidth calculations

Homework 1 Due

Phase 2: Modulation Techniques - Integrated Analog & Digital (Weeks 3-8)

Week 3

Lesson 4 - Tuesday, February 10 Fourier Analysis Applications & Amplitude Modulation Introduction - Spectral analysis of common signals - Parseval's theorem and energy spectral density - Introduction to correlation - Need for modulation - why we modulate signals

READING QUIZ 2 - Topic: Amplitude Modulation Techniques - Reading: Chapter 3.1-3.3 - Administered via Brightspace at beginning of class

Lesson 5 - Thursday, February 12 Amplitude Modulation Theory - DSB-SC (Double Sideband Suppressed Carrier) theory - Full carrier AM - Envelope detection - Modulation index and overmodulation

Homework 2 Assigned: AM and ASK problems

Week 4

Lesson 6 - Tuesday, February 17 Binary Amplitude Shift Keying (ASK) - On-Off Keying (OOK) - Binary ASK as discrete version of AM - Spectral characteristics of ASK - ASK demodulation (coherent and envelope detection) - Superheterodyne receiver and image frequency problem - Comparison of analog AM and digital ASK - Reading: Chapter 7.1-7.2, Chapter 3.6-3.8 (in preparation for Reading Quiz 3 in Lesson 7)

Lesson 7 - Thursday, February 19 SSB Introduction, Python Lab 2, & W3USR Activity 1 - SSB motivation: bandwidth and power efficiency vs. DSB - SSB key formulas (USB, LSB, Hilbert transform overview) - Bandwidth/power comparison: AM vs. DSB-SC vs. SSB - Full SSB treatment (derivation, generation methods, VSB, demodulation) continues in Lesson 8

READING QUIZ 3 - Topic: SSB, VSB, Receiver Architectures, ASK/OOK - Reading: Chapter 3.6-3.8, 7.1-7.2 - Administered via Brightspace at beginning of class

Lab: Python Lab 2 - AM/ASK Modulation - DSB-SC modulation and coherent demodulation - Full carrier AM and envelope detection - OOK modulation and threshold detection - Spectral analysis: AM vs. ASK spectra comparison

Lab Activity: W3USR Station Tour & HF Listening - Tour of W3USR amateur radio station - Listen to SSB voice communications on HF bands (20m, 40m) - Identify AM broadcast signals on 80m band - Demonstrate different receiver modes (AM, SSB, CW)

Homework 2 Due

Week 5

Lesson 8 - Tuesday, February 24 Receiver Architectures - Superheterodyne receiver design - Mixer theory and image frequencies - Direct conversion receivers (produce I/Q baseband) - Direct sampling receivers (modern high-speed ADC approach) - Comparison: superheterodyne vs direct conversion vs direct sampling - Automatic Gain Control (AGC)

Lesson 9 - Thursday, February 26 EXAM 1: Fourier Analysis & Amplitude Modulation - Coverage: Chapters 2-3, 7.1-7.2 - Format: Closed book, equation sheet provided

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Week 6

Lesson 10 - Tuesday, March 3 Exam Review & Angle Modulation Introduction - Exam 1 review and discussion - Introduction to frequency and phase modulation - Narrowband vs. wideband FM - Reading: Chapter 4.1-4.5 (in preparation for Reading Quiz 4 in Lesson 11)

Lesson 11 - Thursday, March 5 FM/PM Theory & Binary FSK - FM/PM mathematical representation - Frequency deviation and modulation index - Narrowband FM approximation - Wideband FM characteristics

READING QUIZ 4 - Topic: FM/PM Theory and Modulation - Reading: Chapter 4.1-4.5 - Administered via Brightspace at beginning of class

Homework 3 Assigned: FM, PM, FSK, BPSK problems

Week 7

Lesson 12 - Tuesday, March 10 FSK Demodulation & Applications - Coherent FSK detection (matched filters / correlators) - Non-coherent FSK detection (envelope detection) - Frequency discriminator for FM/FSK demodulation - Phase-Locked Loop (PLL) for FM demodulation - BER comparison: coherent vs non-coherent FSK vs BPSK - Practical applications: Bell 103 modem, Caller ID, APRS, Bluetooth GFSK - Reading: Chapter 4.6-4.8, 7.3-7.4 (in preparation for Reading Quiz 5 in Lesson 13)

Lesson 13 - Thursday, March 12 BPSK Theory & Demodulation Essentials - Binary Phase Shift Keying fundamentals - BPSK signal generation and constellation - Coherent BPSK detection (correlation receiver, Costas loop overview) - Why BPSK cannot use envelope detection (phase information requires coherent reference) - BER performance of BPSK - Spectral efficiency comparison: ASK vs FSK vs BPSK

READING QUIZ 5 - Topic: FM Generation/Demodulation, FSK/BPSK - Reading: Chapter 4.6-4.8, 7.3-7.4 - Administered via Brightspace at beginning of class

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SPRING BREAK: March 14-22 (No Classes)

Week 8

Lesson 14 - Tuesday, March 24 Homework 3 Review & Baba Yaga Introduction - Return and review Homework 3 solutions (~45 min) - Common mistakes and key takeaways - FM/PM problems, Carson's rule, FSK/MSK, BPSK - Introduction to Baba Yaga's Hut phasor lab (~25 min) - Lab objectives and equipment overview - I/Q demodulator concept and setup - Safety briefing and pair assignments - Reference: w8edu_cwru/the-hut-on-phasors-legs.pdf

Homework 3 Due

Lesson 15 - Thursday, March 26 "Baba Yaga's Hut" - Phasor Analysis Lab (Part 1 of 2) - Full 75-minute lab session - Build I/Q demodulator using function generators and oscilloscopes - Examine AM signals in time, frequency, and phasor domains simultaneously - Vary modulation depth and observe effects in all three representations - Couple to AM radio for audio demodulation - Students work in pairs on the one shared setup - Reference: w8edu_cwru/the-hut-on-phasors-legs.pdf

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Phase 3: Noise, ISI & Advanced Digital (Weeks 9-11)

Week 9

Lesson 16 - Tuesday, March 31 "Baba Yaga's Hut" - Phasor Analysis Lab (Part 2 of 2) - Full 75-minute lab session - Examine DSB-SC (suppressed carrier) signals - Explore FM and PM in phasor representation - Vary carrier phase and observe rotation in phasor domain - Complete lab worksheets and analysis - Students work in pairs on the one shared setup - Reading: Chapter 9 (in preparation for Reading Quiz 6 in Lesson 18)

Thursday, April 2 - NO CLASS (Holy Thursday)

EASTER BREAK: April 3-6 (No Classes)

Week 10

Lesson 17 - Tuesday, April 7 Noise Fundamentals & SNR - Thermal noise generation and statistical properties - White noise and Additive White Gaussian Noise (AWGN) channel model - Noise power spectral density: N /2 - Signal-to-noise ratio (SNR) definition and calculation - Noise figure and noise temperature - Cascaded noise figure (Friis formula) - System noise temperature and receiver design implications - Reading: Chapter 9

Lesson 18 - Thursday, April 9 Link Budgets, ISI & Eye Diagrams + GNU Radio Lab

Part A: Link Budgets & ISI Overview (~30 min) - Friis transmission equation and free-space path loss - EIRP, received power, and link margin - Complete link budget example (satellite or point-to-point) - Intersymbol Interference (ISI) overview: causes, eye diagrams, raised cosine filtering

READING QUIZ 6 - Topic: Noise in Communications Systems - Reading: Chapter 9 - Administered via Brightspace at beginning of class

Part B: GNU Radio Lab 1 - RTL-SDR FM Reception & WiFi Spectrum Analysis (~40 min)

Lab Activity 1: FM Broadcast Reception (20 min) - Introduction to GNU Radio Companion - Connect and configure RTL-SDR (outputs IQ samples) - Build FM broadcast receiver flowgraph: - RTL-SDR source block - Frequency xlating filter to select station - **Limiter block** (remove amplitude variations - Armstrong's contribution!) - WBFM Receive block for demodulation (mono FM, not full stereo/RDS) - Audio sink for output - Visualize waterfall and spectrum displays - Listen to demodulated audio

Lab Activity 2: Understanding FM Demodulation (10 min) - Examine WBFM block internals (discriminator + de-emphasis) - Discuss pre-emphasis/de-emphasis (75 s time constant in US) - Why limiter is critical: removes amplitude noise before FM demod

Lab Activity 3: WiFi Spectrum Analysis (10 min) - Retune RTL-SDR to 2.4 GHz WiFi band - Capture WiFi signals and identify OFDM spectrum shape - Compare OFDM spectrum (flat-topped, rectangular) to single-carrier FM (narrow, peaked)

Week 11

Lesson 19 - Tuesday, April 14 OFDM, Spread Spectrum & CDMA - OFDM fundamentals: subcarrier orthogonality, FFT/IFFT, cyclic prefix - WiFi 802.11a/g OFDM parameters (64 subcarriers, 312.5 kHz spacing, 4 s symbol) - OFDM advantages (multipath robust, adaptive modulation) and disadvantages (PAPR) - Direct Sequence Spread Spectrum (DSSS): processing gain, interference rejection - Frequency Hopping Spread Spectrum (FHSS) overview (Bluetooth: 79 channels, 1600 hops/sec) - PN sequences: m-sequences, Gold codes, autocorrelation properties - CDMA (Code Division Multiple Access) fundamentals - Walsh codes and orthogonality - IQ spreading: separate codes for I and Q channels - Multi-user CDMA: near-far problem, power control - Applications: 3G cellular (CDMA2000, WCDMA), GPS - Comparison: CDMA vs. TDMA vs. FDMA - Reading: Supplemental materials

Lab: Python Lab 3 - CDMA & Spread Spectrum Simulation - Generate and analyze PN sequences

- Simulate DSSS spreading and despreading - Demonstrate processing gain against interference - Multi-user CDMA with Walsh codes - Compare CDMA to TDMA/FDMA

Lesson 20 - Thursday, April 16 M-ary Modulation, QAM, FT8 & W3USR Digital Modes - Extension from binary to M-ary signaling (log (M) bits/symbol) - QPSK: 4 phase states, 2 bits/symbol, Gray coding - Offset QPSK (OQPSK) for satellite links - Higher-order PSK: 8-PSK, 16-PSK limitations - QAM (Quadrature Amplitude Modulation): vary amplitude and phase - 16-QAM, 64-QAM, 256-QAM constellations - Spectral efficiency comparison: BPSK through 256-QAM - WiFi and LTE adaptive modulation (MCS index) - Error Vector Magnitude (EVM): modulation quality metric - EVM specifications: WiFi, LTE, 5G NR requirements - FT8: 8-FSK weak-signal mode, Costas arrays, LDPC coding - Reading: Chapter 7.5-7.7, Supplemental FT8 materials

Lab Activity: W3USR Activity 2 - Digital Modes Demonstration - Observe FT8 operation on HF (20m, 14.074 MHz) using WSJT-X - Identify Costas array tones on waterfall display - Observe SNR reports (negative dB values common) - APRS on 2m VHF (144.390 MHz): 1200 bps AFSK packets - Decode position reports, weather data, messages

Lab: Python Lab 4 - QPSK/QAM Simulation & EVM Analysis - Generate QPSK and QAM constellations - Simulate modulation and demodulation - Add AWGN and observe constellation spreading - Calculate and visualize Error Vector Magnitude (EVM)

Homework 4 Assigned: M-ary modulation, QAM, EVM, spectral efficiency (due Lesson 22)

Phase 4: Exams, Probability & System Performance (Weeks 12-15)

Week 12

Lesson 21 - Tuesday, April 21 EXAM 2: Modulation & Digital Systems - Coverage: Chapters 4-7, 9 (Lessons 10-20) - Topics: FM/PM, FSK, BPSK, noise fundamentals, ISI, OFDM, spread spectrum, QAM, EVM - Format: Closed book, equation sheet provided - Duration: 75 minutes, 100 points

Lesson 22 - Thursday, April 23 Exam Review & Introduction to Probability - Exam 2 review and discussion - Common errors and corrections - Motivation for probability in communications - Random experiments and sample spaces - Probability axioms - Conditional probability and Bayes' theorem - Independent events - Reading: Chapter 8.1-8.2

READING QUIZ 7 - Topic: Probability and Random Variables - Reading: Chapter 8.1-8.2 - Administered via Brightspace at beginning of class

Homework 4 Due

Homework 5 Assigned: Probability, noise, and SNR (due Lesson 26)

Week 13

Lesson 23 - Tuesday, April 28 Probability Fundamentals & Channel Capacity - Conditional probability review: P(A|B) = P(A B) / P(B) - Bayes' theorem and Maximum A Posteriori

(MAP) detection - Worked examples: binary detection, MAP symbol decision

Shannon's Channel Capacity Theorem - Channel capacity definition: maximum reliable data rate - Shannon-Hartley theorem: C = B log (1 + SNR) bits/sec - Spectral efficiency limit: _max = log (1 + SNR) bits/s/Hz - Bandwidth vs. SNR trade-offs - Worked examples: WiFi and LTE capacity calculations

Channel Coding Introduction - Why channel coding: add redundancy for error correction - Simple example: repetition code (rate 1/3) - Modern codes: LDPC, Turbo, Polar (approaching Shannon limit) - Practical system design trade-off example

Lesson 24 - Thursday, April 30 Random Variables & Gaussian Distribution - Discrete and continuous random variables - Probability Mass Functions (PMF) - Probability Density Functions (PDF) - Cumulative Distribution Functions (CDF) - Expected value and variance - Gaussian (Normal) distribution - Q-function and error function - Reading: Chapter 8.3-8.4, Chapter 9 (in preparation for Reading Quiz 8 in Lesson 25)

Week 14

Lesson 25 - Tuesday, May 5 Noise Fundamentals - Thermal noise fundamentals - Noise power spectral density - Additive White Gaussian Noise (AWGN) - Noise figure and noise temperature - SNR calculations for AM and FM systems

READING QUIZ 8 - Topic: Random Variables, Noise in Analog Communications - Reading: Chapter 8.3-8.4, 9 - Administered via Brightspace at beginning of class

Lab: GNU Radio Lab 2 - Noise and SNR Analysis - Add AWGN to AM and FM signals - Measure SNR using spectrum analyzer blocks - Compare noise performance of AM vs. FM - Observe FM threshold effect

Lesson 26 - Thursday, May 7 Data Communication Systems: Sampling & Quantization - Sampling theorem, Nyquist rate, aliasing - Quantization noise and SQNR - Pulse Code Modulation (PCM) - Companding (-law and A-law) - Delta modulation and adaptive delta modulation - Reading: Chapter 10 (in preparation for Reading Quiz 9 in Lesson 27)

Homework 5 Due

Week 15

Lesson 27 - Tuesday, May 12 (LAST CLASS) Digital Performance, Channel Coding & Course Review - Bit Error Rate (BER) fundamentals - BER for BPSK, FSK, and QAM - Performance comparison of modulation schemes - Matched filtering and signal space concepts - Channel coding overview: LDPC (WiFi), Turbo (LTE), Polar (5G) - EVM vs. BER: complementary metrics - Course review and final exam preparation - Q&A session

READING QUIZ 9 - Topic: Digital Performance, BER, System Noise Calculations - Reading: Chapter 10, 11.1-11.3 - Administered via Brightspace at beginning of class

Lab: Python Lab 5 - BER Performance Simulation - Simulate BPSK, QPSK, and QAM in AWGN - Generate BER vs. SNR curves - Compare theoretical and simulated performance

Note: No class Thursday, May 14 (Hamvention) or Tuesday, May 19 (finals week). Final exam is Thursday, May 21.

Final Exam Period

Final Exam - Thursday, May 21, 12:45–2:45 PM

Comprehensive Final Exam - Coverage: All course material (Chapters 1-11) - Emphasis on noise, probability, and system performance (Chapters 8-11) - Format: Closed book, equation sheet provided

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Last updated: March 8, 2026